

Behavioral Responses to Time-Varying Congestion Pricing: A Natural Experiment from New York City

Hitanshu Pandit*

October 2025

[Click here for the latest version](#)

Abstract

Vickrey (1969) bottleneck model predicts that time-varying tolls induce departure-time shifting only where schedule flexibility permits. We test this prediction using New York City's Congestion Relief Zone (CRZ), implemented January 5, 2025, which created a discrete \$6.75 price jump at four pre-announced clock-time boundaries. Applying a difference-in-discontinuities design to hourly mobile-device visit data across Manhattan and Boston establishments, we find behavioral responses at exactly the two boundaries where the Vickrey model predicts them. Overnight establishment visits increase 5.9 percent at the weekday 5 a.m. boundary, where early-arrival costs are modest for shift workers and delivery personnel. Weekend evening establishment activity increases 9.2 percent at the 9 p.m. boundary, where late-arrival costs are low for leisure visitors extending their stay. Estimates at the weekday 9 p.m. and weekend 9 a.m. boundaries are precisely zero, consistent with high institutional penalties at those cutoffs. The asymmetry across boundaries to identical \$6.75 differential, demonstrates that threshold placement relative to schedule technology, not price magnitude, governs congestion pricing effectiveness.

Keywords: Transportation: Demand, Supply, and Congestion; Government Pricing and Policy; Taxation and Subsidies: Externalities; Cross-Sectional Models; Spatial Models; Treatment Effect Models; Quantile Regressions

JEL codes: R41, R48, H23, C21

*Northeastern University, School of Public Policy and Urban Affairs. E-mail: h.pandit@northeastern.edu.

1 Introduction

Vickrey’s (1969) bottleneck model predicts that time-varying tolls generate discontinuous bunching at price thresholds but only where schedule delay costs are low enough that retiming is individually rational. Despite the model’s prominence as the theoretical foundation of congestion pricing, direct empirical tests of this boundary-specific prediction are rare. Evaluations of Stockholm and London congestion pricing model reported average traffic reductions that confound temporal substitution with route diversion, mode switching, and trip suppression (Eliasson et al., 2009; Leape, 2006). These studies reveal that pricing works; they cannot reveal where and why the behavioral mechanism activates.

New York City’s Congestion Relief Zone provides an unusually clean test. From January 5, 2025, vehicles entering Manhattan south of 60th Street pay \$9.00 per trip during peak hours (weekdays 5 a.m. to 9 p.m.; weekends 9 a.m. to 9 p.m.) and \$2.25 off-peak. This creates four fixed, pre-announced toll boundaries, each with an identical \$6.75 differential. The fact that the price change is the same at every cutoff isolates boundary placement and the schedule technology around each cutoff as the sole source of cross-boundary variation in behavioral response.

While Cook et al. (2025), document aggregate traffic and speed effects of the CRZ, their measures cannot distinguish temporal substitution concentrated at toll thresholds from trip suppression and modal shift distributed across the day. We provide the first boundary-specific evidence on hourly temporal substitution. Our contribution is to show not just that the CRZ changes behavior, but precisely which boundaries activate the Vickrey (1969) mechanism and why others do not. This has a direct policy implication, toll threshold placement relative to travelers’ schedule constraints determines whether pricing reduces congestion or merely collects revenue.

We exploit the four-boundary structure using a difference-in-discontinuities design (Grembi et al., 2016) applying to hourly establishment visits, comparing the change in each clock-time discontinuity in Manhattan to the same cutoff in Boston. First, weekday 5 a.m. activity in the congestion zone increased 5.9 percent on the off-peak side of the boundary after implementation consistent with shift workers, delivery personnel, and early-opening retail staff retiming arrivals by one hour to save \$6.75. Second, weekend evening establishment activity increased 9.2 percent at the 9 p.m. boundary consistent with leisure visitors extending stays or arriving later once the toll drops. Two boundaries weekday 9 p.m. and weekend 9 a.m. show precisely zero response,

consistent with binding institutional constraints at those hours. Placebo tests confirm the causal interpretation of both findings.

2 Data and Method

We use mobile-device visit data from Advan, 2025 that are now standard in policy evaluation (Andersen et al., 2023; Coleman et al., 2022; Dave et al., 2021; Goolsbee & Syverson, 2021; Pandit, 2023). We measure hourly visits to points of interest from July 2023 through July 2025. Visits are attributed to the hour bin in which they begin; a device entering an establishment at 8:47 p.m. and remaining until 10:20 p.m. generates one count in the 8 p.m. bin only. Any observed change in a post-cutoff hour bin therefore reflects visit starts in that window, not lingering presence from earlier arrivals. The unit of observation is establishment-level visit starts, not zone-level unique persons: a visitor stopping at a restaurant and then a bar after 9 p.m. generates two post-cutoff counts. This distinction is inconsequential at the weekday 5 a.m. boundary, where pre-dawn activity is sparse and single-destination. At the weekend 9 p.m. boundary the estimate captures a combination of new zone entries and multi-stop activity by existing visitors, which we acknowledge in interpreting that result. The balanced panel requires each establishment to appear in every week of the sample, yielding 24,704 Manhattan and 15,234 Boston establishments. Institutionally rigid sectors (healthcare, government, religious organizations, air transportation) are excluded. Table 1 confirms near identical baseline log-visit means (10.47 in Manhattan, 10.52 in Boston). The identifying assumption requires only that discontinuity sizes at toll boundaries would have evolved similarly in Manhattan and Boston absent the policy a weaker condition than parallel trends in levels which Figure A1 and the placebo date test in Table A4 support.

Table 1: Sample Composition and Baseline Activity

Sector (NAICS)	Manhattan (CRZ)		Boston (Control)	
	<i>N</i>	Mean	<i>N</i>	Mean
Retail (441–453)	6,244	10.82	4,112	11.04
Services (522–611)	9,135	11.19	5,317	11.22
Food & Accommodation	4,285	11.87	2,443	12.14
Other	5,040	9.43	3,362	9.67
Total	24,704	10.47	15,234	10.52

Notes: Mean = log visitors, baseline period (calendar year 2024). *N* is the number of establishments. Full sector-by-geography table in Online Appendix A1.

The core challenge is that establishment visits exhibit natural discontinuities at salient clock times business openings, shift changes, commuting rhythms that exist in both cities regardless of toll pricing. A simple regression discontinuity would confound these baseline jumps with any pricing response. We implement the difference-in-discontinuities design of Grembi et al. (2016), which tests whether the *size* of the discontinuity changed differentially in Manhattan relative to Boston after implementation:

$$\begin{aligned}
Y_{ight} = \exp(&\alpha_t + \gamma_g + \rho_0 + \rho_1 r_h + \rho_2 r_h \cdot D_h^O \\
&+ \beta_1 D_h^O + \beta_2 D_g^M \cdot P_t + \beta_3 D_h^O \cdot P_t + \beta_4 D_g^M \cdot D_h^O \\
&+ \tau \cdot D_g^M \cdot D_h^O \cdot P_t) + \varepsilon_{igt}
\end{aligned} \tag{1}$$

where Y_{ight} is visits to establishment i in geography g , hour h , date t ; r_h measures hours from the toll cutoff (running variable); $D_h^O = \mathbf{1}[r_h < 0]$ flags the off-peak (low-toll) side; D_g^M flags Manhattan; P_t flags post-January 5, 2025 dates; α_t are date fixed effects; and γ_{gh} are city-by-hour fixed effects absorbing pre-existing differences in hourly activity profiles between cities non-parametrically. The parameter of interest, τ , is the difference-in-discontinuities estimand capturing the change in the size of the toll-boundary discontinuity in Manhattan relative to Boston after implementation. We estimate Equation (1) by Poisson pseudo-maximum likelihood (PPML), separately for each boundary, clustering standard errors by date.

3 Results

Table 2, reports the main estimates. The parameter of interest is $\hat{\tau}$ (Manhattan $\times$ Off-Peak $\times$ Post). The triple-interaction estimate is 0.058, implying a 5.9 percent increase in off-peak visits in the congestion zone relative to Boston after implementation. The Off-Peak $\times$ Post coefficient (-0.146^{***}) reveals that the overnight-versus-daytime contrast shifted sharply in both cities during this period a differential seasonal or return-to-office shock that would have badly confounded a simple RD comparison. The Manhattan $\times$ Post coefficient (0.049^{**}) shows that overall early-morning activity rose in the treatment area, a level shift absorbed by the triple-difference structure. The residual $\hat{\tau}$ captures only the additional discontinuity shift at the toll threshold itself.

Table 2: Difference-in-Discontinuities Estimates by Toll Boundary

	Weekday		Weekend	
	5 A.M.	9 P.M.	9 A.M.	9 P.M.
Manhattan $\times$ Post	0.0485*** (0.0173)	0.0196 (0.0177)	-0.0054 (0.0249)	0.0032 (0.0289)
Off-Peak $\times$ Post	-0.1464*** (0.0243)	0.0227 (0.0296)	0.0175 (0.0253)	0.0208 (0.0592)
Manhattan $\times$ Off-Peak $\times$ Post	0.0575** (0.0268)	-0.0253 (0.0356)	0.0083 (0.0332)	0.0878** (0.0321)
Date FE	Yes	Yes	Yes	Yes
City $\times$ Hour FE	Yes	Yes	Yes	Yes
Observations	7,634	7,634	3,036	3,036
Pseudo R^2	0.967	0.955	0.971	0.940

Notes: PPML estimates. Manhattan = 1 for visitor in store located in CRZ area, Off-Peak = 1 on the low-toll side of each cutoff, Post = 1 after January 5, 2025. Standard errors clustered by date in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The economic mechanism is transparent. At 5 a.m., avoiding the peak toll requires arriving *earlier*. For shift workers, delivery personnel, and early-opening retail staff, the marginal schedule-delay cost (β) of shifting a trip from 5:30 a.m. to 4:30 a.m. is modest relative to the \$6.75 saving, satisfying the Vickrey condition $\Delta\tau > \beta \cdot \Delta t$. Sectoral heterogeneity in Online Appendix C is consistent with this interpretation: consumer-facing retail and banking show positive point estimates at this boundary, while logistics sectors (couriers, food manufacturing) show near-zero estimates consistent with coordination constraints that bind regardless of price.

At 9 P.M. on weekends, the triple-interaction estimate is 0.088 (s.e. = 0.032), implying a 9.2 percent increase in establishment visits in the off-peak window after the toll drops at 9 p.m. Two important clarifications govern how this estimate should be interpreted. First, the behavioral mechanism differs from the 5 a.m. case. At 9 p.m. on weekends, avoiding the peak toll requires delaying trips staying in the zone longer or arriving after the boundary. The relevant schedule-delay parameter is late-arrival cost (γ), which for weekend leisure activity is substantially lower than on weekday evenings. Weekend visitors face no childcare cutoff, no next-day coordination constraint, and no business-closure deadline; leisure activity dining, entertainment, socializing extends naturally past 9 p.m. at low cost. This is precisely the condition under which the Vickrey model predicts temporal substitution: $\Delta\tau > \gamma \cdot \Delta t$ is satisfied when γ is low. Second, because the data measure establishment-level unique visitors rather than zone-level unique persons, this estimate captures a combination of two behavioral responses that

cannot be separated with available data: new visitors entering the zone after 9 p.m., and existing visitors extending their stay and visiting additional establishments once the toll drops. Both responses are economically meaningful the first is pure temporal substitution in the Vickrey sense, the second is activity intensification driven by the same price signal but they have distinct implications for congestion relief. This limitation matters less for the 5 a.m. boundary where pre-dawn activity is sparse and single-destination, making multi-stop amplification implausible.

At weekday 9 p.m. and weekend 9 a.m. boundaries, both estimates are small, signed inconsistently, and statistically indistinguishable from zero. At weekday 9 p.m., avoiding the toll requires delaying trips past 9 p.m. on a work night. Late-arrival penalties are prohibitive: childcare ends, morning commutes begin in hours, coordination with others fails. The null is precisely what the model predicts when γ is high. At weekend 9 a.m., avoiding the toll requires arriving before 9 a.m. earlier than the leisure schedules that dominate weekend mornings. Fixed social commitments (brunch reservations, events, coordination with groups) make early-arrival retiming infeasible regardless of the price signal. Two significant and two null estimates with significance concentrated at the two boundaries where the model predicts effects is not a multiple-testing concern but a pattern Vickrey (1969)’s theory specifies.

Table 3, reports placebo cutoff tests shifting each assumed boundary by one hour. At the weekday 5 a.m. boundary, the triple-interaction estimate is 0.058 ($p < 0.05$) at the actual cutoff, shrinks to 0.030 (insignificant) at 4 a.m., and collapses to -0.001 at 6 a.m. The near-zero at 6 a.m. confirms the knife-edge pattern: no discontinuity change at the post-cutoff placebo hour. The positive but insignificant 4 a.m. estimate reflects behavioral spillover from toll avoiders who shift arrivals to one hour before the cutoff, not one hour after an asymmetric pattern that is itself predicted by the retiming mechanism.

The weekend 9 p.m. placebo pattern warrants careful interpretation. The pre-cutoff (8 p.m.) estimate is 0.023, only 26 percent of the actual boundary estimate and cleanly insignificant, ruling out a pre-existing upward trend in weekend evening activity in Manhattan as an explanation for the main result. The post-cutoff (10 p.m.) estimate is 0.050 (57 percent of the main estimate), narrowly insignificant. Critically, this asymmetry between placebo estimates small before the cutoff, moderate after is consistent with the behavioral mechanism. At 9 p.m. the toll drops, so the predicted behavioral response is delayed activity, which should appear on the post-cutoff side. Visitors who shift their timing do not arrive precisely at 9:01 p.m.; they land disproportionately in the 10 p.m. to midnight window as dwell-time extension and later arrivals

Table 3: Placebo Cutoff Tests

	Weekday Morning		
	One Hour Earlier	Actual Boundary	One Hour Later
Time tested	4:00 a.m.	5:00 a.m.	6:00 a.m.
Manhattan $\times$ Off-Peak $\times$ Post S.E.	0.0302 (0.0449)	0.0575** (0.0268)	-0.0007 (0.0204)
Date FE	Yes	Yes	Yes
City $\times$ Hour FE	Yes	Yes	Yes
Observations	7,634	7,634	7,634
	Weekend Evening		
	One Hour Earlier	Actual Boundary	One Hour Later
Time tested	8:00 p.m.	9:00 p.m.	10:00 p.m.
Manhattan $\times$ Off-Peak $\times$ Post S.E.	0.0232 (0.0301)	0.0878** (0.0321)	0.0498 (0.0341)
Date FE	Yes	Yes	Yes
City $\times$ Hour FE	Yes	Yes	Yes
Observations	3,036	3,036	3,036

Notes: PPML estimates. Manhattan = 1 for visitor in store located in CRZ area. Off-Peak = 1 on the low-toll side of each cutoff. Post = 1 after January 5, 2025. Standard errors clustered by date in parentheses.

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

spread across the evening. This is why the 5 a.m. placebo test produces a sharp knife-edge (retiming to an exact departure time is feasible) while the 9 p.m. test produces an asymmetric pattern (extending an evening is inherently diffuse). The pre-cutoff placebo cleanly excludes confounding; the elevated post-cutoff placebo reflects the predicted character of the behavioral response, not its absence.

A placebo implementation date of September 5, 2024 provides a second validity check in Table A4. The weekday 5 A.M. and 9 P.M. triple-interaction coefficient falls and all boundaries, placebo-date estimates become insignificant, ruling out pre-existing differential trends between Manhattan and Boston as an explanation for either primary finding.

4 Conclusion

These results refine the Vickrey bottleneck model's empirical implications. Time-varying tolls generate discontinuous temporal substitution at boundaries where schedule flexibility is present, and precisely zero response at boundaries where institutional constraints bind even when the

price differential is identical. Two boundaries produce significant effects, the weekday 5 a.m. cutoff, where low early-arrival costs enable retiming by shift workers and delivery personnel, and the weekend 9 p.m. cutoff, where low late-arrival costs enable dwell-time extension and delayed arrivals by leisure visitors. However, two boundaries produce zero effects, the weekday 9 p.m. cutoff, where late-arrival penalties are prohibitive on work nights, and the weekend 9 a.m. cutoff, where fixed social commitments constrain early-morning retiming.

The finding that the weekend 9 p.m. response is smoother and less knife-edge than the 5 a.m. response is itself theoretically informative. Early-arrival retiming admits a focal departure time; dwell-time extension and late-arrival substitution are inherently spread across hours. The nature of the behavioral response sharp or diffuse reflects the nature of the schedule-flexibility mechanism, not the strength of the price signal. For policy perspective, boundary placement should be mapped against the schedule-flexibility landscape of the affected population. Early-morning thresholds targeting workers with flexible start times generate measurable departure-time shifting. When temporal substitution is constrained as at institutional boundaries congestion relief must operate through mode switching, trip suppression, or longer-run location adjustment, each with distinct welfare and distributional implications.

This study captures short-term responses to the NYC-CRZ. Moreover, establishment visits proxy for economic activity but do not directly measure vehicle entries or congestion. The analysis cannot separate new zone entrants from dwell-time extension at the weekend 9 p.m. boundary. These caveats aside, the four-boundary structure of the CRZ provides unusually direct evidence that the Vickrey mechanism is real, boundary-specific, and governed by schedule technology rather than price alone.

Acknowledgments

The author thanks Martin S. Andersen, Mindy Marks, and Alicia Modestino for helpful comments.

Disclosure Statement: Use of AI Tools

Portions of this manuscript were prepared with the assistance of OpenAI’s ChatGPT (GPT-5) to improve clarity, structure, and conciseness of the writing. All data analysis, interpretation of results, and substantive conclusions are the author’s own. The AI tool was not used to generate,

alter, or verify empirical results or citations. The author reviewed and edited all AI-assisted text to ensure accuracy, originality, and compliance with academic integrity standards.

Disclosure of Interest

The author report there are no competing interests to declare.

Disclosure of Funding

The author received funding under IHS-Grant-IHS019702 for access to the Advan Research data .

References

- Advan, R. (2025). Foot traffic / weekly patterns plus [dataset]. <https://doi.org/10.82551/C103-N851>
- Andersen, M., Maclean, J. C., Pesko, M. F., & Simon, K. (2023). Does paid sick leave encourage staying at home? evidence from the united states during a pandemic. *Health Economics*, 32(6), 1256–1283.
- Coleman, N., Gao, X., DeLeon, J., & Mostafavi, A. (2022). Human activity and mobility data reveal disparities in exposure risk reduction indicators among socially vulnerable populations during covid-19 for five us metropolitan cities. *Scientific Reports*, 12(1), 15814.
- Cook, C., Kreidieh, A., Vasserman, S., Allcott, H., Arora, N., van Sambeek, F., Tomkins, A., & Turkel, E. (2025). *The short-run effects of congestion pricing in new york city* (tech. rep.). National Bureau of Economic Research.
- Dave, D., Friedson, A. I., Matsuzawa, K., & Sabia, J. J. (2021). When do shelter-in-place orders fight covid-19 best? policy heterogeneity across states and adoption time. *Economic inquiry*, 59(1), 29–52.
- Eliasson, J., Hultkrantz, L., Nerhagen, L., & Rosqvist, L. S. (2009). The stockholm congestion–charging trial 2006: Overview of effects. *Transportation Research Part A: Policy and Practice*, 43(3), 240–250.
- Goolsbee, A., & Syverson, C. (2021). Fear, lockdown, and diversion: Comparing drivers of pandemic economic decline 2020. *Journal of public economics*, 193, 104311.

- Grembi, V., Nannicini, T., & Troiano, U. (2016). Do fiscal rules matter? *American Economic Journal: Applied Economics*.
- Leape, J. (2006). The london congestion charge. *Journal of Economic Perspectives*, 20(4), 157–176.
- Pandit, H. (2023). *Mobile-device location data and labor markets*. The University of North Carolina at Greensboro.
- Vickrey, W. S. (1969). Congestion theory and transport investment. *American Economic Review*.